INFORMATION RETRIEVAL

ADVANCED INFORMATION RETRIEVAL

Week 4: Probabilistic and Structured Retrieval Models



Topics

- Probabilistic Retrieval Model
- Structured Text Retrieval
- Overlapped Lists
- Non-Overlapped Lists
- Proximal Nodes

Week 4

Learning Outcomes

By the end of **Week 4**, students will be able to:

1. Explain probabilistic retrieval.
2. Understand the Probability Ranking Principle.
3. Implement probability-based ranking.
4. Explain structured text retrieval.
5. Differentiate overlapped and non-overlapped lists.
6. Use proximal nodes for structure-based retrieval.
7. Connect these models with FYP, research, product, and funding outcomes.



What Is Probabilistic Retrieval?

Probabilistic Retrieval estimates the probability that a document is relevant to a query.

Instead of saying only “match” or “no match,” it says:

“This document is likely to be relevant.”

The system ranks documents according to their **probability of relevance.**



Why Probabilistic Retrieval Is Important

Boolean Model
gives exact matching.

Vector Space Model
gives similarity.

Probabilistic Model
gives likelihood of relevance.



It is useful when:

- User queries are **uncertain**
- Documents have **partial matches**
- Ranking needs **statistical support**
- Search systems need **relevance estimation**

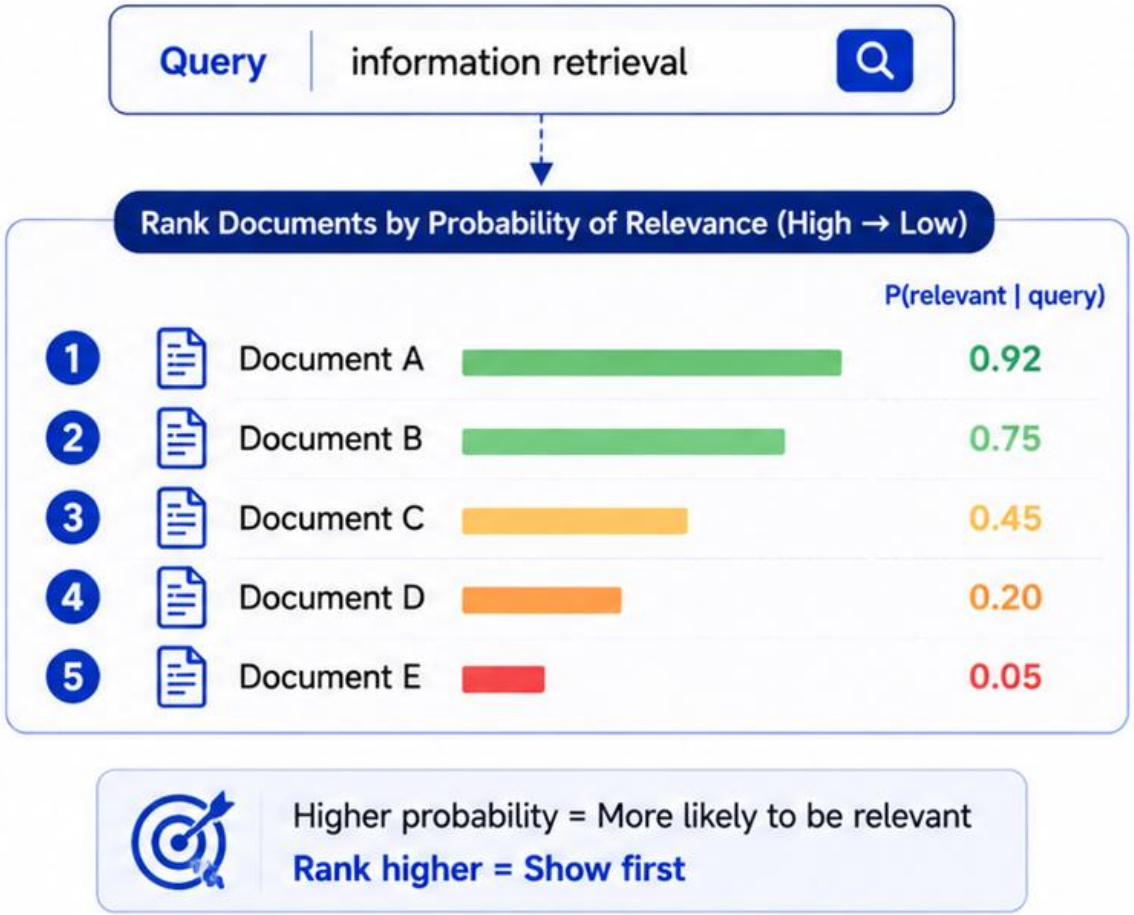
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Probability Ranking Principle

The **Probability Ranking Principle** states that documents should be ranked in **decreasing order** of their probability of relevance to the user query.

In simple words:

The document most likely to satisfy the user should appear first.



Higher probability = More likely to be relevant

Rank higher = Show first

Week 4

Mathematical Expression

A simple **probability-based ranking** expression can be: $P(\text{Relevance} \mid Q, D)$

This means:

Probability that **document D** is relevant given **query Q**.

In simple words:

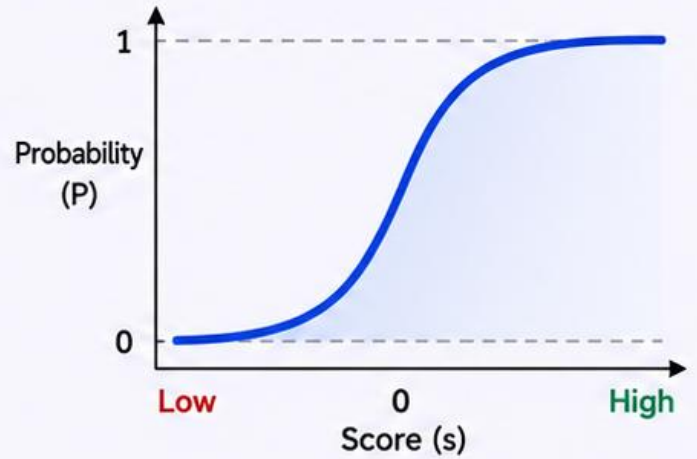
Estimate how likely a document matches the user's need.

Scoring to Probability Conversion

$$P = \frac{1}{1 + e^{-s}}$$

- P = probability of relevance
- s = score between query and document
- e = natural exponential constant

How it Works



Higher score (s) → **Higher probability** (P)

Week 4

Binary Independence Model

The Binary Independence Model represents terms as present or absent.

If a term is present: 1

If a term is absent: 0

Example: Query: cybersecurity attack detection

A document is checked for whether these terms are present or absent.

The more useful term evidence it contains, the more likely it is relevant.

Query Terms	Documents			
	D1	D2	D3	D4
cybersecurity	1	1	0	1
attack	1	0	1	0
detection	1	1	0	1

The more useful term evidence a document contains, the more likely it is relevant.

BIM uses **yes/no term presence as input**, but combines those binary signals with **probabilities to rank documents by relevance**.

BM25 as Practical Probabilistic Ranking

BM25 is a widely used practical ranking model.

It improves simple term matching by considering:

- Term **frequency**
- Document **length**
- Collection-level term **importance**
- Query-document **matching**



cybersecurity attack detection

BM25 Ranking Results

		Score
1	Document C (Best match)	2.13
2	Document A (Good match)	1.48
3	Document B (Fair match)	0.92

For **IR development**, **BM25** can be used directly in **FYP search modules**.

For **Advanced IR**, **BM25** can be studied, compared, improved, or extended for research.

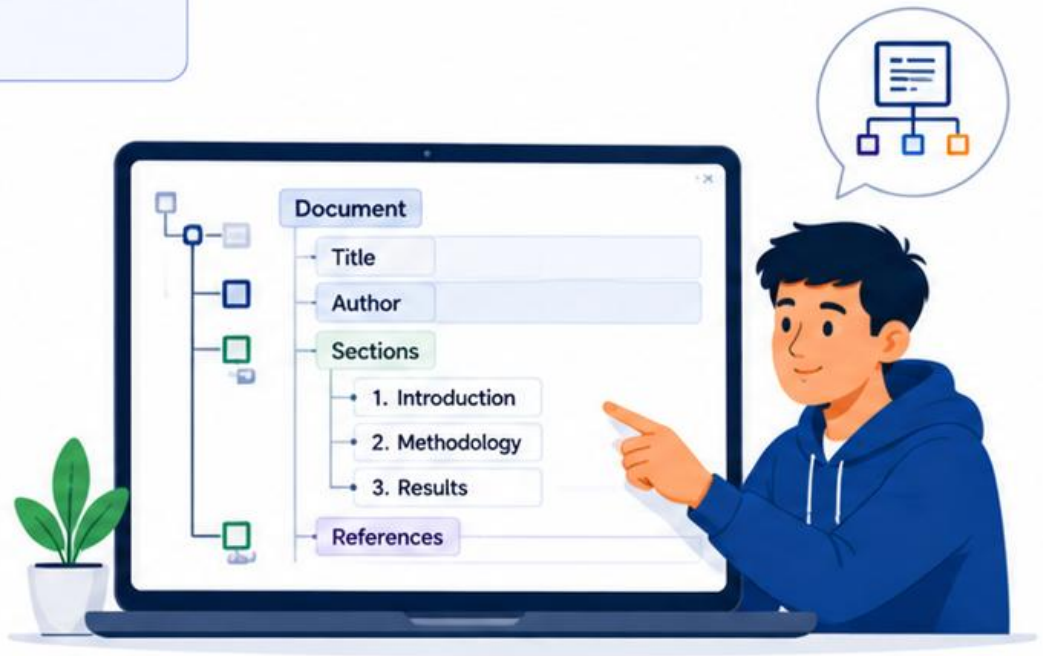
BM25 is like a smarter BIM: it ranks documents by checking whether query words appear, how often they appear, and how long the document is.

What Is Structured Text Retrieval?

Structured Text Retrieval retrieves information from documents that have **internal structure**.

Examples

- **XML** documents
- **JSON** files
- **Databases**
- **Research** papers
- **Legal** documents
- **Books with** chapters and sections
- **Product** catalogues



Structured retrieval does not treat the document as one flat text. It considers **fields**, **sections**, **nodes**, and **regions**.

Fields | Sections | Nodes | Regions

Key Concepts in Structured Text Retrieval

Match Point:

The position where query terms match the document text.

Example

Document

Cybersecurity systems are essential for detecting and preventing cyber attacks. Effective attack detection helps protect sensitive data and maintain system integrity.

Match Points

Query: cyber attack detection

Region:

A continuous part of text, such as a paragraph or section.

Example

1. Introduction

2. Attack Detection

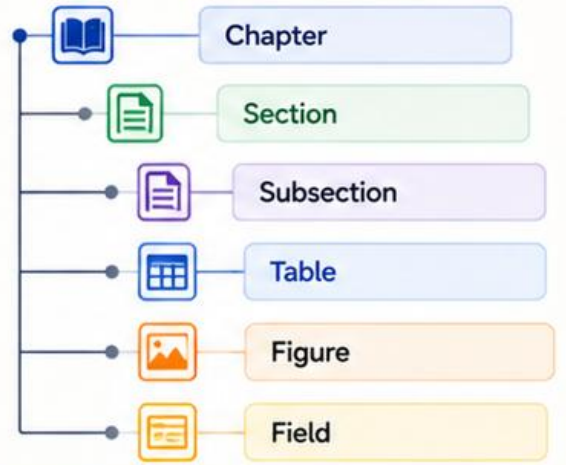
3. Related Work

Region

Node:

A structural component, such as chapter, section, subsection, table, figure, or field.

Example



Structured retrieval understands the structure of documents and uses match points, regions, and nodes to find more accurate and relevant information.

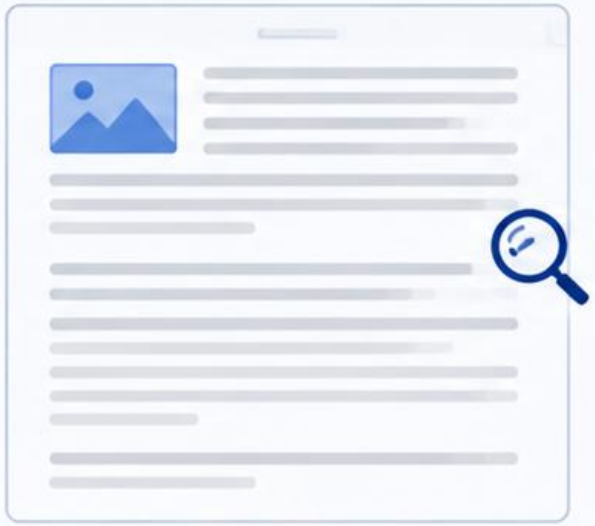
Example of Structured Retrieval

Query:
"Pakistani Flag in image caption"



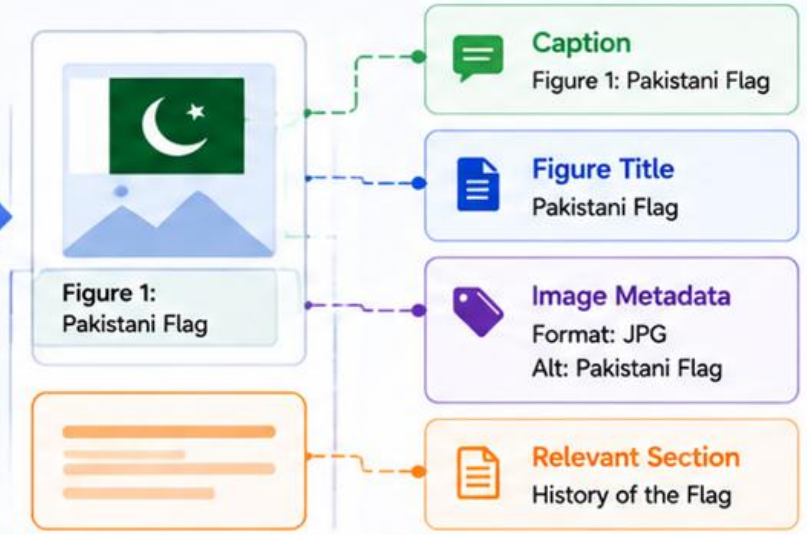
Normal Text Retrieval

A normal text retrieval system may search the entire document.



Structured Retrieval

A structured retrieval system can search only specific parts.



Structured Retrieval Can Search Only:

- Captions
- Figure titles
- Image metadata
- Relevant sections

This makes retrieval **more precise.**

Overlapped Lists Model

The **Overlapped Lists Model** retrieves documents or regions where multiple query terms overlap or occur together.

Example

Query:

cats AND dogs

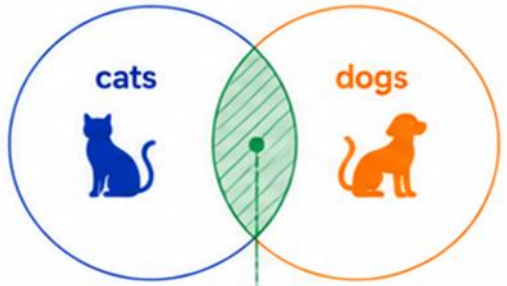
The system searches for documents or regions where both terms occur together.

This improves **precision**.

Posting Lists (Documents)

List for "cats"		List for "dogs"	
D1	1	D1	1
D2	1	D2	0
D3	1	D3	0
D4	1	D4	1
D5	1	D5	1

Overlap (cats AND dogs)



Documents/Regions where **BOTH** terms occur together

Results (Higher Precision)

1. D1 — Cats and dogs together
2. D4 — Cats and dogs together
3. D5 — Cats and dogs together

Non-Overlapped Lists Model

The Non-Overlapped Lists Model divides text into **non-overlapping regions**.

Each region belongs to a **list**.

The system retrieves regions/documents where terms **appear separately**, even if they do not occur in the same exact region.

This **improves coverage** but may **reduce precision**.

Example: Non-Overlapped Lists

List: cybersecurity	List: attack	List: detection
R1 (Section 1)	R2 (Section 1)	R3 (Section 2)
R3 (Section 2)	R4 (Section 3)	R4 (Section 3)
R5 (Section 4)	R6 (Section 5)	R7 (Section 6)

Query: cybersecurity attack detection

→ **Retrieved Regions/Documents: R1, R2, R3, R4, R5, R6, R7**
(Terms may appear in different regions/lists.)



Example Document Divided into Non-Overlapping Regions

Region ID	Region Content (Non-Overlapping)
R1 (Section 1)	**Cybersecurity** is essential for protecting systems and data.
R2 (Section 1)	**Attackers** exploit vulnerabilities to gain unauthorized access.
R3 (Section 2)	Effective **cybersecurity** monitoring helps in early **detection**.
R4 (Section 3)	Advanced **attack detection** techniques use AI and analytics.
R5 (Section 4)	**Cybersecurity** awareness reduces the risk of cyber threats.

How Retrieval Works

- Each term is searched in its own list.
- Regions from all lists are combined.
- Documents or regions are retrieved even if terms are in different regions.

Key Insight

- **Higher coverage** (finds more relevant documents)
- **Lower precision** (includes some less relevant documents)

Overlapped vs Non-Overlapped Lists

These two models represent **different ways** to organize term occurrences for retrieval.

Feature	Overlapped Lists	Non-Overlapped Lists
Main Idea	Terms occur together	Terms may occur separately
Strength	Higher **precision**	Higher **coverage**
Weakness	May **miss** broader results	May return **irrelevant** results
Best Use	**Exact** multi-term search	**Exploratory** structured search

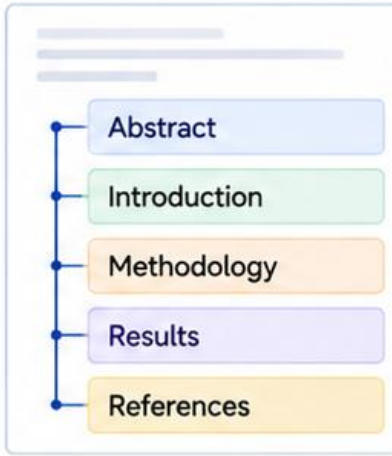
Proximal Nodes Model

The Proximal Nodes Model retrieves information by considering **nearby** or **connected** structural nodes.

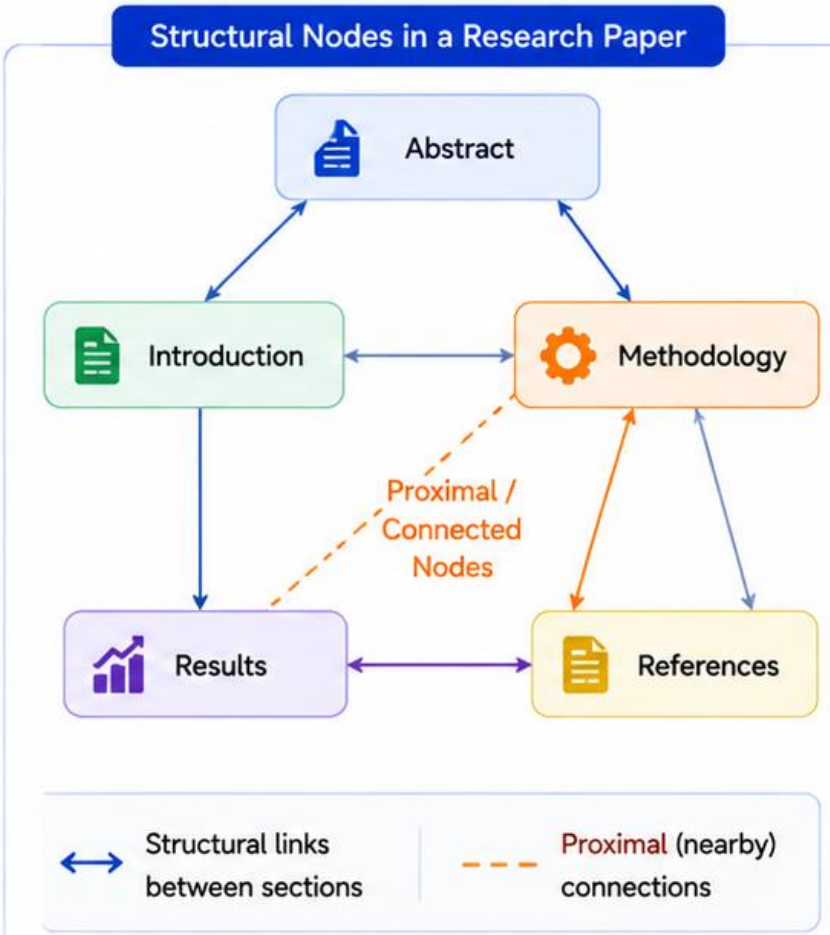
Example

In a research paper:

- Abstract
- Introduction
- Methodology
- Results
- References



If the query term appears in **Methodology** and related terms appear in **Results**, both nodes may be treated as **connected** and useful.





IR Development Focus

For IR students, Week 4 must result in a **practical module**:

1. Use **probability-based ranking**.
2. Search **structured fields**.
3. Retrieve **document regions**.
4. Support **exact** and **broader** matching.
5. Show **ranked results** in a user interface.

What the Module Should Do

User Query
(e.g., AI in healthcare)

→ **Search Structured Fields**
(Title, Abstract, Methodology, Results)

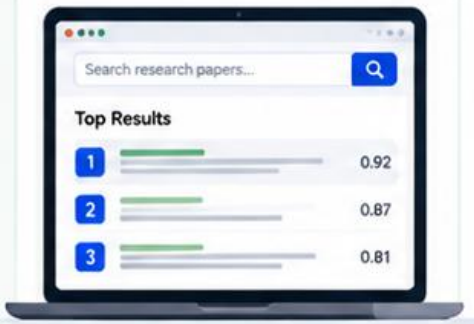
→ **Retrieve Relevant Regions**
(Match in specific sections)

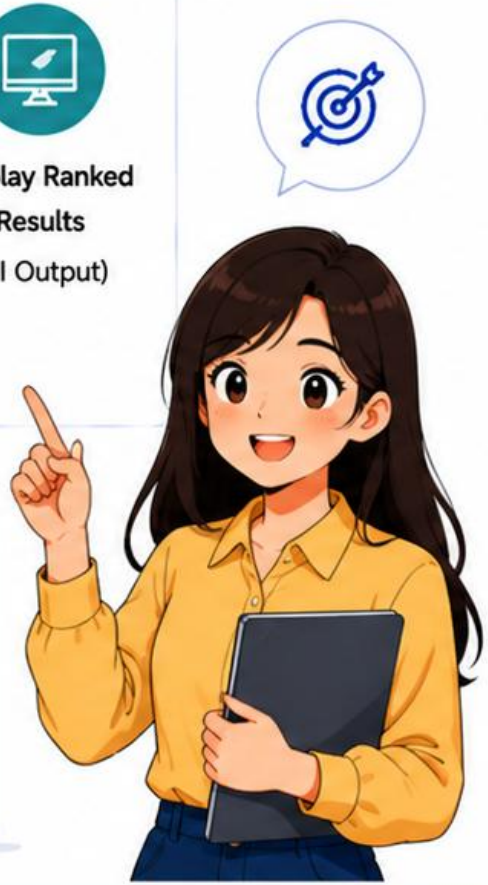
→ **Rank Using Probability Model**
(BM25 / Probabilistic)

→ **Display Ranked Results**
(UI Output)

Example FYP Module

"Search within **research papers** by **title**, **abstract**, **methodology**, and **results**."





Advanced IR Research/Product Focus

For Advanced IR students, Week 4 must result in:

1. Mathematical probability-based ranking.
2. Structured document representation.
3. Node-region modelling.
4. Research gap in structured retrieval.
5. Product idea using field-level ranking.
6. Paper/funding proposal based on structured or probabilistic retrieval.

Key Research Areas

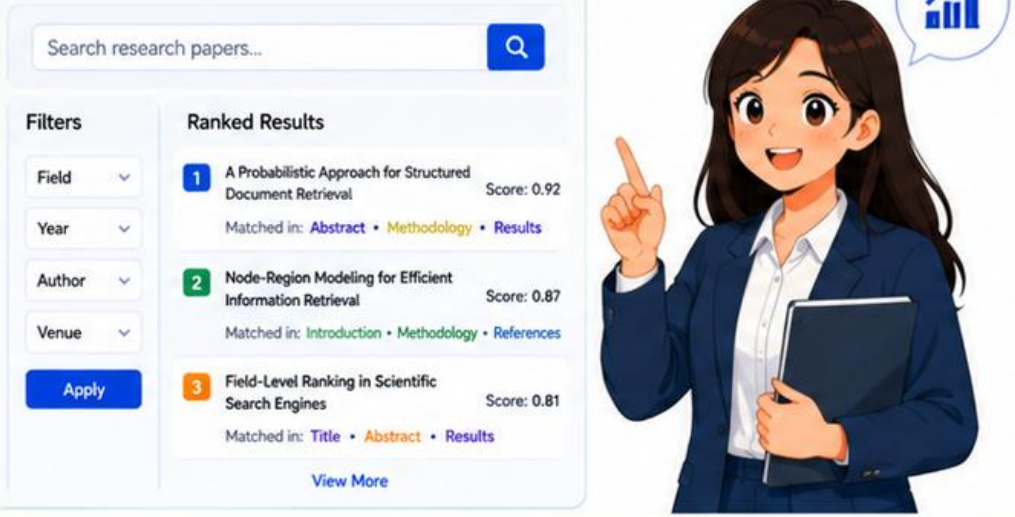
- **Probabilistic Models** (BM25, Language Models, etc.)
- **Structured Representation** (Fields, Sections, Metadata)
- **Node-Region Models** (Proximal nodes, Non-overlapped lists, etc.)
- **Research Gaps** (Efficiency, Accuracy, Explainability, Hybrid Approaches)
- **Product Ideas** (Field-aware search, Smart ranking, AI integration)
- **Papers / Funding Proposals** (Novel methods, Datasets, Impact)

Product Idea Example

Smart Research Paper Search

An intelligent search system that:

- Searches across title, abstract, methodology, results, and references.
- Ranks using probabilistic field-level scores.
- Shows results with matched sections highlighted.
- Supports filters, faceted search and recommendations.



Case Study Assignment

Week 4 assignment focuses on applying IR models for **practical development** and advancing **research/product ideas**.

For IR Students

Implement retrieval by:

1. **Probabilistic ranking**
   (e.g., BM25 or other probability models)
2. **Structured text search**
   (e.g., fields like title, abstract, methodology, results)
3. **Non-overlapped list search**
   (retrieve from separate regions/lists)
4. **Proximal node-based search**
   (retrieve using nearby/connected nodes)

Deliverable: Working system with modules and results.

For Advanced IR Students

Prepare a **research/product proposal** that mathematically explains:

Probability-based ranking

- Mathematical formulation (e.g., BM25, language models)
- Term weighting, normalization, scoring
- Why it improves retrieval effectiveness

Structure-aware retrieval

- Document fields and regions
- Node/region models (non-overlapped, proximal nodes, etc.)
- How structure improves relevance

Deliverable: Research paper / Product proposal with mathematical models and evaluation plan.



Quiz

Answer the following questions based on **Week 4** concepts.

1. What is the main idea of probabilistic retrieval?
2. What does $P(Relevance \mid Q, D)$ mean?
3. How does structured retrieval differ from unstructured retrieval?
4. What is the difference between a region and a node?
5. Which model gives better precision: overlapped or non-overlapped lists?
6. How can proximal nodes be useful in research paper retrieval?



Summary

Week 4 covered:

- Probabilistic Retrieval Model
- Probability Ranking Principle
- Binary Independence Model
- BM25 idea
- Structured Text Retrieval
- Match points, regions, and nodes
- Overlapped Lists
- Non-Overlapped Lists
- Proximal Nodes
- FYP/product implementation
- Research and mathematical modelling direction